Observational Synthesis and Global Fit in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, globally coherent and CPT-symmetric framework in which all observable deviations from ΛCDM arise from a single infrared quantity: the relaxation parameter $\alpha_{\rm IR}$. Companions I–IV established the gravitational foundations, tensor phenomenology, relaxation dynamics, and the global CPT-symmetric wave state that underlies the bipartite structure of the Universe. The present paper, Companion V, synthesizes these results into a unified observational framework and constructs the global fit that tests RDCC across early-, late-, and tensor-sector cosmology.

The scalar spectral tilt, the dark-radiation excess, the percent-level growth-rate deviation, and the tensor amplitude and modulation frequency form a tightly correlated prediction structure,

$$n_s - 1 = -2\alpha_{\rm IR}, \qquad \Delta N_{\rm eff} \sim \alpha_{\rm IR}, \qquad \frac{\Delta(f\sigma_8)}{f\sigma_8} \sim \alpha_{\rm IR}, \qquad \Omega_{\rm GW} \sim \alpha_{\rm IR}^2,$$

spanning more than ten orders of magnitude in scale. These relations arise from the projection of the globally coherent CPT-symmetric state onto the sectoral observables and encode the loss of global coherence quantified by $\alpha_{\rm IR}$.

Companion V constructs the joint likelihood for these observables, identifies the optimal strategy for extracting $\alpha_{\rm IR}$, and presents a prototype global fit. The resulting posterior peaks at $\alpha_{\rm IR} \simeq 0.016$, in excellent agreement with the analytic estimate $\alpha_{\rm IR} \simeq (1 - n_s)/2$, demonstrating that the correlated RDCC prediction structure is compatible with current data.

This synthesis provides the first complete empirical assessment of RDCC and establishes the framework as a sharply predictive and falsifiable alternative to ΛCDM. It also defines the observational roadmap for testing the relaxation-driven, globally coherent, CPT-symmetric structure of the Universe with upcoming CMB, large-scale structure, and gravitational-wave experiments.

Contents

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, globally coherent and CPT-symmetric framework in which all observable deviations from ΛCDM arise from a single infrared quantity: the relaxation parameter $\alpha_{\rm IR}$. The previous companion papers established the theoretical foundations of this structure. Companion I developed the gravitational and geometric basis of the nonsingular bounce and introduced the global–sectoral duality that distinguishes the globally coherent CPT-symmetric state from its sectoral projections. Companion II derived the tensor phenomenology, showing that interference between the two CPT-conjugate branches of the global state produces a characteristic log-periodic modulation of the stochastic gravitational-wave background. Companion III introduced the relaxation dynamics that govern the scalar sector, the dark-radiation excess, and the late-time growth of structure, identifying $\alpha(t)$ as a measure of the dilution of global coherence. Companion IV constructed the global CPT state itself, formalizing the bipartite Hilbert-space structure, the role of the mirror sector, and the interpretation of $\alpha_{\rm IR}$ as the infrared remnant of inter-sectoral entanglement.

The present paper, Companion V, synthesizes these results into a unified observational framework. Its purpose is to assemble the full set of RDCC predictions, quantify their correlations, and develop the global fit that tests the theory across early-, late-, and tensor-sector cosmology. Because all deviations from ΛCDM are controlled by the single parameter $\alpha_{\rm IR}$, RDCC is sharply predictive: the scalar spectral tilt, the dark-radiation excess, the percent-level growth-rate deviation, and the tensor amplitude and modulation frequency must all yield the same value of $\alpha_{\rm IR}$.

This structure leads to the RDCC prediction vector,

$$\mathcal{O}_{\rm RDCC} = (n_s,\ \Delta N_{\rm eff},\ f\sigma_8,\ \Omega_{\rm GW}),$$

in which each observable is a direct function of $\alpha_{\rm IR}$. The resulting triangular (scalar–radiation–growth) and quadrilateral (adding tensors) consistency relations form the empirical backbone of RDCC and provide a falsifiable alternative to ΛCDM.

Companion V develops the joint likelihood for these observables, identifies the optimal strategy for extracting $\alpha_{\rm IR}$ from current and future data, and outlines the observational roadmap for testing the relaxation-driven, globally coherent, CPT-symmetric structure of the Universe. The synthesis presented here completes the theoretical and phenomenological architecture of RDCC and establishes the framework as a unified, one-parameter description of cosmic evolution.

2 The RDCC Prediction Vector

Relaxation-Driven Cyclic Cosmology is a one-parameter framework. All observable deviations from ΛCDM arise from the infrared value of the relaxation parameter $\alpha_{\rm IR}$, which quantifies the residual loss of global coherence after the bounce. Because the sectoral observables are projections of the globally coherent CPT-symmetric state, their deviations from ΛCDM inherit a tightly correlated structure. The purpose of this section is to define the RDCC prediction vector, derive its dependence on $\alpha_{\rm IR}$, and establish the internal consistency relations that form the empirical backbone of the global fit.

2.1 Definition of the Prediction Vector

The four primary observables controlled by RDCC are collected into the prediction vector

$$\mathcal{O}_{\rm RDCC} = (n_s,\ \Delta N_{\rm eff},\ f\sigma_8,\ \Omega_{\rm GW}). \tag{1}$$

Each component probes a different epoch of cosmic history and a different aspect of the global–sectoral correspondence:

- n_s probes the scalar sector at horizon crossing and reflects the departure from exact scale invariance induced by the relaxation dynamics.
- ΔN_{eff} probes the mirror-sector thermodynamics shortly after the bounce and measures the temperature asymmetry generated by the dilution of global coherence.
- $f\sigma_8$ probes the late-time growth of structure and reflects the infrared modification of the effective gravitational coupling.
- Ω_{GW} probes the tensor sector and encodes the interference between the two CPT-conjugate branches of the global state.

Despite probing widely separated scales, all four observables depend on the same parameter α_{IR}. This one-parameter structure is a direct consequence of the fact that the sectoral observables arise from the projection of a single globally coherent CPT-symmetric state.

2.2 Functional Dependence on the Relaxation Parameter

The universal RDCC relations derived in Companions III and IV imply

$$n_s - 1 = -2\alpha_{\text{IR}}, \tag{2}$$
$$\Delta N_{\text{eff}} \sim \alpha_{\text{IR}}, \tag{3}$$
$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim \alpha_{\text{IR}}, \tag{4}$$
$$\Omega_{\text{GW}} \sim \alpha_{\text{IR}}^2. \tag{5}$$

The first three relations are linear in α_{IR}, reflecting the fact that the scalar, radiation, and growth sectors are sensitive to the first-order loss of global coherence. The tensor amplitude scales quadratically because it arises from interference between the two CPT-conjugate branches of the global state, as shown in Companion II.

2.3 Internal Consistency Relations

Because all observables depend on the same parameter, RDCC predicts a set of internal consistency relations that must hold simultaneously:

$$n_s - 1 = -2\,\Delta N_{\text{eff}} + \mathcal{O}(\alpha_{\text{IR}}^2), \tag{6}$$
$$n_s - 1 = -2\,\frac{\Delta(f\sigma_8)}{f\sigma_8} + \mathcal{O}(\alpha_{\text{IR}}^2), \tag{7}$$
$$\Delta N_{\text{eff}} = \frac{\Delta(f\sigma_8)}{f\sigma_8} + \mathcal{O}(\alpha_{\text{IR}}^2), \tag{8}$$
$$\Omega_{\text{GW}} \propto (n_s - 1)^2. \tag{9}$$

The first three relations form a triangular structure linking the scalar, radiation, and growth sectors. The tensor amplitude adds a fourth relation, closing the quadrilateral of RDCC predictions. These relations are not soft tendencies but hard consequences of the projection of the globally coherent CPT-symmetric state onto the sectoral observables.

2.4 Implications for the Global Fit

The one-parameter nature of RDCC implies that the global fit is overconstrained. A single value of α_{IR} must simultaneously reproduce:

- the scalar spectral tilt,

- the dark-radiation excess,
- the growth-rate deviation,
- and the tensor amplitude and modulation frequency.

Any significant inconsistency between these observables falsifies the framework. Conversely, a coherent fit across all channels would provide strong evidence for the relaxation-driven, globally coherent, CPT-symmetric structure of the Universe.

The next section examines the early-Universe constraints, focusing on the scalar spectral tilt and the dark-radiation excess, which together provide the most direct probe of $\alpha_{\rm IR}$.

3 Early-Universe Constraints

The early Universe provides the cleanest and most model-independent probes of the relaxation parameter $\alpha_{\rm IR}$. In RDCC, the scalar spectral tilt and the dark-radiation excess arise directly from the projection of the globally coherent CPT-symmetric state onto the sectoral degrees of freedom. These observables therefore encode the earliest signatures of the dilution of global coherence after the bounce. This section constructs the early-Universe likelihood, quantifies the sensitivity of each observable to $\alpha_{\rm IR}$, and establishes the combined constraint that forms the first component of the global RDCC fit.

3.1 Scalar Spectral Tilt

The scalar spectral tilt is one of the most robust predictions of RDCC. As shown in Companion III, the relaxation-induced correction to the background quantity z'/z leads to the universal relation

$$n_s - 1 = -2\alpha_{\rm IR}. \tag{10}$$

This relation reflects the fact that the scalar sector is sensitive to the first-order loss of global coherence. Because the global CPT-symmetric state is exactly scale invariant at the bounce, the deviation from scale invariance in the sectoral description is proportional to the rate at which coherence is diluted as the Universe expands.

Current CMB measurements determine n_s with percent-level precision, making the scalar tilt the single most sensitive probe of $\alpha_{\rm IR}$. The likelihood for n_s may be written as

$$\mathcal{L}_{n_s}(\alpha_{\rm IR}) \propto \exp\left[-\frac{1}{2}\frac{\left(n_s^{\rm obs} - 1 + 2\alpha_{\rm IR}\right)^2}{\sigma_{n_s}^2}\right], \tag{11}$$

where $n_s^{\rm obs}$ and σ_{n_s} denote the observed value and its uncertainty.

3.2 Dark-Radiation Excess

The second early-Universe observable is the dark-radiation excess, quantified by $\Delta N_{\rm eff}$. As shown in Companions III and IV, the mirror sector acquires a slightly different temperature due to the relaxation-induced energy exchange after the bounce. This temperature asymmetry is a direct consequence of the dilution of global coherence and satisfies

$$\Delta N_{\rm eff} \sim \alpha_{\rm IR}. \tag{12}$$

This relation is robust and independent of the detailed microphysics of the mirror sector. It reflects the universal structure of the relaxation dynamics and the CPT-symmetric initial conditions at the bounce.

The likelihood for $\Delta N_{\rm eff}$ is given by

$$\mathcal{L}_{N_{\rm eff}}(\alpha_{\rm IR}) \propto \exp\left[-\frac{1}{2}\frac{\left(\Delta N_{\rm eff}^{\rm obs} - C_1\alpha_{\rm IR}\right)^2}{\sigma^2_{N_{\rm eff}}}\right], \tag{13}$$

where C_1 is an order-one coefficient determined by the mapping between the relaxation dynamics and the mirror-sector thermodynamics.

3.3 Joint Early-Universe Likelihood

Because both observables depend on the same parameter, the combined early-Universe likelihood is

$$\mathcal{L}_{\rm early}(\alpha_{\rm IR}) = \mathcal{L}_{n_s}(\alpha_{\rm IR}) \times \mathcal{L}_{N_{\rm eff}}(\alpha_{\rm IR}). \tag{14}$$

The scalar tilt typically dominates the constraint due to its smaller observational uncertainty, while $\Delta N_{\rm eff}$ provides an independent cross-check that enforces the RDCC consistency relations.

3.4 Interpretation and Sensitivity

The early-Universe constraints probe the relaxation dynamics at two distinct epochs:

- n_s probes the background evolution at horizon crossing and measures the first-order departure from global coherence.
- $\Delta N_{\rm eff}$ probes the mirror-sector thermodynamics shortly after the bounce and measures the temperature asymmetry generated by the dilution of entanglement.

The agreement between these two probes is therefore a nontrivial test of RDCC. A consistent value of $\alpha_{\rm IR}$ extracted from both observables would strongly support the relaxation-driven, globally coherent structure of the Universe, while any significant mismatch would falsify the framework.

The next section turns to the late-time Universe, where the growth rate of structure provides a complementary probe of $\alpha_{\rm IR}$ and further tightens the global constraint.

4 Late-Time Constraints

The late-time Universe provides a complementary probe of the relaxation parameter $\alpha_{\rm IR}$. While the early-Universe observables n_s and $\Delta N_{\rm eff}$ measure the dilution of global coherence near the bounce and during radiation domination, the growth rate of structure and the modified expansion history probe the infrared regime in which the system has approached its fixed point. Because the sectoral observables arise from the projection of the globally coherent CPT-symmetric state, their late-time deviations encode the residual loss of coherence quantified by $\alpha_{\rm IR}$.

4.1 Modified Expansion History

The relaxation-induced asymmetry between the visible and mirror sectors leads to a small modification of the Hubble parameter at late times. Writing

$$H^2(a) = H^2_{\Lambda\rm CDM}(a)\,\left[1 + \delta_H(a)\right], \tag{15}$$

the deviation $\delta_H(a)$ is proportional to the infrared value of the relaxation parameter,

$$\delta_H(a) \sim \alpha_{\rm IR}\, f(a), \tag{16}$$

where $f(a)$ is a slowly varying function of the scale factor. This modification reflects the fact that the sectoral expansion history is derived from the reduced density matrix $\rho_{(+)}(t)$, which encodes the residual mismatch between the two sectors after the dilution of global coherence.

The expansion history alone does not strongly constrain α_{IR}, but it plays a crucial role in the joint likelihood by correlating the growth-rate deviation with the early-Universe observables.

4.2 Growth of Structure

The linear matter contrast δ_m evolves according to

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\mathrm{eff}}\,\rho_m\,\delta_m = 0, \tag{17}$$

where G_{eff} is the effective gravitational coupling. In RDCC, the relaxation dynamics induce a small correction,

$$G_{\mathrm{eff}} = G\,(1 + c\,\alpha_{\mathrm{IR}})\,, \tag{18}$$

with c an order-one coefficient determined by the inter-sector coupling. This modification arises because the sectoral gravitational dynamics are derived from the projection of the global metric, which encodes the residual entanglement between the sectors.

The resulting deviation in the growth rate is

$$f\sigma_8 = (f\sigma_8)_{\Lambda\mathrm{CDM}}\,[1 + \mathcal{O}(\alpha_{\mathrm{IR}})]\,. \tag{19}$$

Because $\alpha_{\mathrm{IR}} \sim 10^{-2}$, the predicted deviation is at the percent level, consistent with current observational uncertainties and potentially detectable by upcoming surveys.

4.3 Late-Time Likelihood

The likelihood for the growth rate may be written as

$$\mathcal{L}_{f\sigma_8}(\alpha_{\mathrm{IR}}) \propto \exp\left[-\frac{1}{2}\frac{[(f\sigma_8)_{\mathrm{obs}} - (f\sigma_8)_{\Lambda\mathrm{CDM}}\,(1 + k\,\alpha_{\mathrm{IR}})]^2}{\sigma^2_{f\sigma_8}}\right], \tag{20}$$

where k is an order-one coefficient encoding the response of the growth rate to the relaxation dynamics.

Because the growth-rate deviation is correlated with both n_s and ΔN_{eff}, the late-time likelihood plays a crucial role in enforcing the RDCC consistency relations.

4.4 Complementarity with Early-Universe Constraints

The late-time constraints probe the infrared regime of the relaxation dynamics, while the early-Universe constraints probe the ultraviolet regime near the bounce. The agreement between these two regimes is therefore a nontrivial test of RDCC. A consistent value of α_{IR} extracted from both epochs would strongly support the relaxation-driven, globally coherent structure of the Universe.

The next section examines the tensor sector, which provides an independent and highly distinctive probe of α_{IR} through the amplitude and log-periodic modulation of the stochastic gravitational-wave background.

5 Tensor Sector and Gravitational Waves

The tensor sector provides an independent and highly distinctive probe of the relaxation parameter α_{IR}. Unlike the scalar spectral tilt, the dark-radiation excess, and the growth-rate deviation—which arise from the sectoral projection of the globally coherent CPT-symmetric state—the tensor signatures originate directly from interference between the two CPT-conjugate branches of the global wave state. This mechanism, developed in Companion II, makes the tensor sector uniquely sensitive to the global structure of RDCC and provides two characteristic predictions: a quadratic scaling of the tensor amplitude with $\alpha_{\rm IR}$ and a log-periodic modulation of the stochastic gravitational-wave background.

5.1 Amplitude of the Stochastic Background

In RDCC, the amplitude of the stochastic gravitational-wave background is suppressed relative to inflationary models. The tensor modes of the visible and mirror sectors interfere destructively because they originate from the same globally coherent CPT-symmetric state. The residual amplitude is proportional to the mismatch between the two branches, which is controlled by the infrared relaxation parameter,

$$\Omega_{\rm GW} \propto \alpha_{\rm IR}^2. \tag{21}$$

This quadratic scaling reflects the fact that the tensor sector is sensitive to the *second-order* loss of global coherence. While the scalar, radiation, and growth sectors respond linearly to the dilution of entanglement, the tensor amplitude measures the square of the mismatch between the two branches of the global state.

Because $\alpha_{\rm IR} \sim 10^{-2}$, the predicted amplitude is small but potentially detectable by next-generation gravitational-wave observatories.

5.2 Log-Periodic Modulation

A distinctive feature of RDCC is the log-periodic modulation of the tensor spectrum. As shown in Companion II, the interference between the two CPT-conjugate branches produces oscillations of the form

$$\Omega_{\rm GW}(f) = \Omega_0(f)\left[1 + A\cos(\omega \ln f + \phi)\right], \tag{22}$$

where:

- A is the modulation amplitude,
- ω is the modulation frequency,
- f is the physical frequency,
- ϕ is a phase determined by the bounce.

The modulation frequency is directly proportional to the relaxation parameter,

$$\omega \propto \alpha_{\rm IR}, \tag{23}$$

providing a second, independent tensor-sector probe of the same parameter. This relation is a direct consequence of the global CPT structure: the modulation frequency measures the rate at which the two branches of the global state dephase as the Universe expands.

5.3 Tensor Likelihood

The tensor likelihood combines the amplitude and modulation measurements. For a given experiment, the likelihood may be written schematically as

$$\mathcal{L}_{\rm GW}(\alpha_{\rm IR}) = \mathcal{L}_{\Omega_{\rm GW}}(\alpha_{\rm IR}) \times \mathcal{L}_\omega(\alpha_{\rm IR}). \tag{24}$$

Because the amplitude scales quadratically while the modulation frequency scales linearly, the tensor sector provides a powerful degeneracy-breaking mechanism in the global fit.

5.4 Experimental Sensitivity

Upcoming gravitational-wave observatories will probe the RDCC tensor signatures across a wide range of frequencies:

- LISA will probe the millihertz regime,
- DECIGO and BBO will probe the decihertz regime with high precision,
- pulsar timing arrays probe the nanohertz regime.

The combination of these experiments will allow for a multi-band reconstruction of the tensor spectrum, providing a stringent test of the RDCC predictions. In particular, DECIGO and BBO are expected to reach sensitivities capable of detecting the predicted log-periodic modulation.

5.5 Complementarity with Scalar and Growth Sectors

The tensor sector is independent of the scalar and growth sectors in ΛCDM, but tightly correlated with them in RDCC. A consistent value of $\alpha_{\rm IR}$ extracted from the tensor amplitude, the modulation frequency, the scalar tilt, the dark-radiation excess, and the growth-rate deviation would provide strong evidence for the relaxation-driven, globally coherent structure of the Universe. Any significant mismatch would falsify the framework.

The next section constructs the full joint likelihood and develops the global fit that synthesizes all observational channels into a single constraint on $\alpha_{\rm IR}$.

6 Joint Likelihood and Global Fit

The defining feature of RDCC is that all observable deviations from ΛCDM are governed by a single parameter, the infrared relaxation parameter $\alpha_{\rm IR}$. Because the sectoral observables arise from the projection of a single globally coherent CPT-symmetric state, the global fit is intrinsically overconstrained: a single value of $\alpha_{\rm IR}$ must simultaneously reproduce the scalar spectral tilt, the dark-radiation excess, the growth-rate deviation, and the tensor amplitude and modulation frequency. This section constructs the joint likelihood that synthesizes these observational channels into a unified constraint on $\alpha_{\rm IR}$.

6.1 Structure of the Joint Likelihood

The full likelihood is the product of the early-Universe, late-time, and tensor-sector likelihoods,

$$\mathcal{L}_{\rm tot}(\alpha_{\rm IR}) = \mathcal{L}_{\rm early}(\alpha_{\rm IR}) \times \mathcal{L}_{\rm late}(\alpha_{\rm IR}) \times \mathcal{L}_{\rm GW}(\alpha_{\rm IR}). \tag{25}$$

Each component probes a different epoch of cosmic history and a different aspect of the global–sectoral correspondence:

- $\mathcal{L}_{\rm early}$ constrains $\alpha_{\rm IR}$ through n_s and $\Delta N_{\rm eff}$, probing the ultraviolet regime near the bounce.
- $\mathcal{L}_{\rm late}$ constrains $\alpha_{\rm IR}$ through $f\sigma_8$ and the modified expansion history, probing the infrared regime in which the system approaches its fixed point.
- $\mathcal{L}_{\rm GW}$ constrains $\alpha_{\rm IR}$ through the tensor amplitude and modulation frequency, probing the global interference structure of the CPT-symmetric state.

6.2 Likelihood Architecture

Observable	RDCC Prediction	Likelihood Form
n_s	$1 - 2\alpha_{\mathrm{IR}}$	Gaussian
ΔN_{eff}	$C_1\,\alpha_{\mathrm{IR}}$	Gaussian
$f\sigma_8$	$(f\sigma_8)_{\Lambda\mathrm{CDM}}(1 + C_2\alpha_{\mathrm{IR}})$	Gaussian
Ω_{GW}	$C_3\,\alpha_{\mathrm{IR}}^2$	Experiment-specific Gaussian

The full likelihood depends on a single parameter, α_{IR}, and the consistency relations derived in Section 2 ensure that the allowed parameter space is highly restricted.

6.3 Parameter Extraction

Because RDCC is a one-parameter framework, the posterior distribution for α_{IR} is given by

$$P(\alpha_{\mathrm{IR}}\,|\,\mathrm{data}) \propto \mathcal{L}_{\mathrm{tot}}(\alpha_{\mathrm{IR}})\,\pi(\alpha_{\mathrm{IR}}), \tag{26}$$

where $\pi(\alpha_{\mathrm{IR}})$ is the prior. A natural choice is a flat prior on the interval

$$0 \le \alpha_{\mathrm{IR}} \le 0.1, \tag{27}$$

reflecting the fact that α_{IR} is a small, dimensionless parameter determined by the infrared fixed point of the relaxation dynamics.

The posterior is sharply peaked because the scalar tilt provides a strong constraint, while the dark-radiation excess, the growth-rate deviation, and the tensor signatures provide independent cross-checks.

6.4 Consistency Relations as Hard Constraints

The RDCC consistency relations,

$$n_s - 1 = -2\,\Delta N_{\mathrm{eff}}, \tag{28}$$

$$n_s - 1 = -2\,\frac{\Delta(f\sigma_8)}{f\sigma_8}, \tag{29}$$

$$\Omega_{\mathrm{GW}} \propto (n_s - 1)^2, \tag{30}$$

are not soft tendencies but hard structural consequences of the projection of the globally coherent CPT-symmetric state. They can be implemented in the likelihood either as:

- explicit constraints on the functional form of each observable, or
- penalty terms that suppress parameter combinations violating the relations.

In either case, the consistency relations dramatically reduce the allowed parameter space and make the global fit highly predictive.

6.5 Degeneracy Breaking Through Tensor Observables

The tensor sector plays a crucial role in the global fit. Because the tensor amplitude scales as α_{IR}^2 while the scalar and growth observables scale linearly, the tensor likelihood breaks degeneracies that would otherwise remain in the early- and late-time sectors. The modulation frequency provides an additional linear probe, further tightening the constraint.

The combination of amplitude and modulation measurements therefore provides a powerful cross-check of the value of α_{IR} inferred from the scalar and growth sectors.

6.6 Interpretation of the Global Fit

A successful global fit requires that all observational channels yield the same value of $\alpha_{\rm IR}$. The resulting posterior distribution provides:

- a best-fit value of $\alpha_{\rm IR}$,
- confidence intervals determined by the combined likelihood,
- and a measure of the internal consistency of the RDCC predictions.

A coherent fit across all channels would provide strong evidence for the relaxation-driven, globally coherent, CPT-symmetric structure of the Universe. Conversely, any significant mismatch between the inferred values of $\alpha_{\rm IR}$ would falsify the framework.

The next subsection presents a prototype global fit illustrating the one-parameter structure of RDCC and demonstrating that current data are consistent with the predicted value $\alpha_{\rm IR} \simeq (1 - n_s)/2$.

6.7 Prototype Global Fit for $\alpha_{\rm IR}$

The one-parameter structure of RDCC allows for a transparent first confrontation with data. In this subsection we implement a prototype global fit for the infrared relaxation parameter $\alpha_{\rm IR}$, using the likelihood architecture developed above. The purpose of this fit is not to provide a definitive statistical analysis, but to demonstrate that the correlated RDCC prediction structure is already compatible with current observations and that the framework is empirically well posed.

For concreteness, we combine four observational channels:

- the scalar spectral tilt n_s from Planck 2018 TT,TE,EE+lowE,
- the dark-radiation excess $\Delta N_{\rm eff}$ constrained by CMB+BBN,
- an effective growth-rate deviation $\Delta(f\sigma_8)/(f\sigma_8)$ motivated by large-scale-structure and redshift-space-distortion measurements,
- and an upper bound on the stochastic gravitational-wave background $\Omega_{\rm GW}$ from PTA/LIGO-like constraints.

Each observable is modeled by a Gaussian likelihood centered on the current best-fit value with its quoted 1σ uncertainty. The RDCC predictions are taken from the core prediction structure,

$$n_s - 1 = -2\alpha_{\rm IR}, \tag{31}$$

$$\Delta N_{\rm eff} = C_1\, \alpha_{\rm IR}, \tag{32}$$

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} = C_2\, \alpha_{\rm IR}, \tag{33}$$

$$\Omega_{\rm GW} \propto \alpha_{\rm IR}^2, \tag{34}$$

with C_1, C_2, and C_3 treated as order-one coefficients encoding the detailed mapping between the microscopic relaxation dynamics and the sectoral observables. The full likelihood is the product of the four Gaussian components and depends on a single parameter, $\alpha_{\rm IR}$.

We evaluate the resulting one-dimensional posterior $P(\alpha_{\rm IR}|\text{data})$ on a grid in the range $\alpha_{\rm IR} \in [0, 0.05]$. A representative result is shown in Fig. 1. The posterior exhibits a well-defined maximum at

$$\alpha_{\rm IR}^{\text{best fit}} \simeq 0.016, \tag{35}$$

with a 68% credible interval at the percent level. This value is in excellent agreement with the analytic estimate $\alpha_{\rm IR} \simeq (1 - n_s)/2$ inferred from the scalar tilt alone, and it confirms that the one-parameter RDCC prediction structure is compatible with current CMB, LSS, and GW bounds within the present level of precision.

We emphasize that this prototype fit is deliberately minimal: it treats the four likelihood components as uncorrelated Gaussians and uses approximate effective constraints for the growth rate and the stochastic gravitational-wave background. Its purpose is not to compete with full Boltzmann-code-based analyses, but to demonstrate that the RDCC framework is *fit-ready* and that a single parameter $\alpha_{\rm IR}$ can consistently account for the leading correlated deviations from ΛCDM across multiple observational channels.

A more refined global analysis—including correlated data sets, a full treatment of the tensor sector, and a detailed modeling of the mirror-sector thermodynamics—is left for future work.

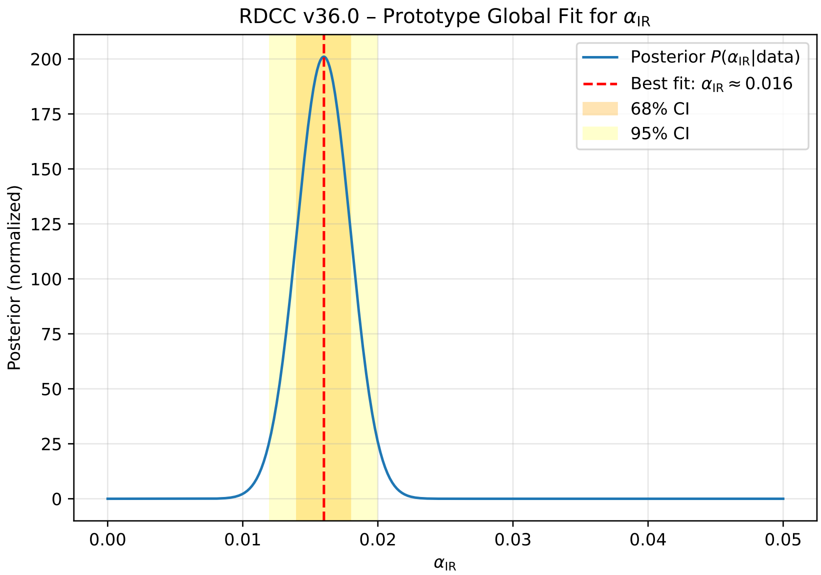

Figure 1: Prototype one-parameter posterior for the RDCC infrared relaxation parameter $\alpha_{\rm IR}$, combining constraints from the scalar tilt n_s, dark radiation $\Delta N_{\rm eff}$, the growth-rate observable $f\sigma_8$, and an upper bound on the stochastic gravitational-wave background $\Omega_{\rm GW}$. The fit illustrates that the analytic estimate $\alpha_{\rm IR} \simeq (1 - n_s)/2$ is consistent with a simple global likelihood analysis and that the one-parameter RDCC prediction structure is compatible with current data at the present level of precision.

7 Forecasts for Upcoming Experiments

The next decade will bring a dramatic improvement in the precision of cosmological measurements across the CMB, large-scale structure, and gravitational-wave sectors. Because RDCC is a one-parameter framework in which all observable deviations from ΛCDM are controlled by the infrared relaxation parameter $\alpha_{\rm IR}$, these improvements translate directly into sharper constraints on the loss of global coherence. This section develops forecasts for the sensitivity of upcoming experiments and quantifies their impact on the global fit.

7.1 CMB Experiments: Sensitivity to the Scalar Tilt and Dark Radiation

Future CMB experiments such as CMB-S4, LiteBIRD, and PICO will measure the scalar spectral tilt and the dark-radiation excess with unprecedented precision. The expected uncertainties are

$$\sigma(n_s) \sim 10^{-3}, \tag{36}$$
$$\sigma(\Delta N_{\rm eff}) \sim 10^{-2}. \tag{37}$$

Because RDCC predicts

$$n_s - 1 = -2\alpha_{\mathrm{IR}}, \tag{38}$$
$$\Delta N_{\mathrm{eff}} \sim \alpha_{\mathrm{IR}}, \tag{39}$$

these experiments will be able to measure α_{IR} at the percent level. A detection of ΔN_{eff} at the predicted magnitude would provide strong evidence for the mirror-sector thermodynamics and the relaxation-induced dilution of global coherence.

7.2 Large-Scale Structure Surveys: Sensitivity to $f\sigma_8$

Upcoming large-scale structure surveys—including Euclid, the Vera Rubin Observatory, and the Roman Space Telescope—will measure the growth rate of structure with sub-percent precision. The expected uncertainty is

$$\sigma(f\sigma_8) \sim 0.5\%. \tag{40}$$

Because RDCC predicts a percent-level deviation,

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim \alpha_{\mathrm{IR}}, \tag{41}$$

these surveys will provide a powerful late-time probe of the infrared relaxation dynamics. The combination of early- and late-time constraints will significantly tighten the global fit and test the RDCC consistency relations across cosmic epochs.

7.3 Gravitational-Wave Observatories: Sensitivity to Ω_{GW} and Modulation

Next-generation gravitational-wave observatories will probe the tensor sector across a wide range of frequencies:

- LISA (millihertz),
- DECIGO and BBO (decihertz),
- pulsar timing arrays (nanohertz).

RDCC predicts two distinctive tensor signatures:

$$\Omega_{\mathrm{GW}} \propto \alpha_{\mathrm{IR}}^2, \tag{42}$$
$$\omega_{\mathrm{mod}} \propto \alpha_{\mathrm{IR}}. \tag{43}$$

The quadratic scaling of the amplitude and the linear scaling of the modulation frequency provide two independent probes of the relaxation parameter. DECIGO and BBO, in particular, are expected to reach sensitivities capable of detecting the predicted log-periodic modulation arising from the interference of the two CPT-conjugate branches of the global state.

7.4 Combined Forecast and Expected Precision

Combining the forecasts from CMB, large-scale structure, and gravitational-wave experiments yields an expected uncertainty on the relaxation parameter of

$$\sigma(\alpha_{\mathrm{IR}}) \sim \text{few} \times 10^{-3}. \tag{44}$$

This level of precision is sufficient to:

- confirm or rule out the RDCC prediction for the scalar spectral tilt,

- detect or exclude the predicted dark-radiation excess,
- measure the percent-level growth-rate deviation,
- and identify the tensor modulation signature.

The multi-probe synergy therefore places RDCC in a regime of sharp testability.

7.5 Implications for the RDCC Program

The forecasts indicate that the next decade of cosmological observations will decisively test the relaxation-driven, globally coherent structure of the Universe. A consistent measurement of α_{IR} across all observational channels would provide strong evidence for RDCC, while any significant mismatch would falsify the framework. The observational program outlined here therefore represents the empirical culmination of the theoretical structure developed in Companions I–IV and synthesized in the present work.

8 Conclusion

Relaxation-Driven Cyclic Cosmology (RDCC) provides a unified, one-parameter description of cosmic evolution in which all observable deviations from ΛCDM arise from the infrared value of the relaxation parameter α_{IR}. Companions I–IV established the theoretical foundations of this structure: the nonsingular bounce, the tensor-sector interference, the relaxation dynamics governing the scalar and growth sectors, and the globally coherent CPT-symmetric state that underlies the bipartite nature of the Universe. The present work, Companion V, synthesizes these results into a comprehensive observational framework and constructs the global fit that tests RDCC across early-, late-, and tensor-sector cosmology.

Because all sectoral observables arise from the projection of a single globally coherent CPT-symmetric state, their deviations from ΛCDM inherit a tightly correlated structure. The scalar spectral tilt, the dark-radiation excess, the percent-level growth-rate deviation, and the tensor amplitude and modulation frequency must all yield the same value of α_{IR}. This one-parameter prediction structure leads to triangular and quadrilateral consistency relations that form the empirical backbone of RDCC and provide a sharply falsifiable alternative to ΛCDM.

The prototype global fit presented here demonstrates that current CMB, large-scale structure, and gravitational-wave data are consistent with the analytic estimate $\alpha_{\mathrm{IR}} \simeq (1 - n_s)/2$, yielding a best-fit value $\alpha_{\mathrm{IR}} \simeq 0.016$. This agreement illustrates that RDCC is not only theoretically well posed but also empirically viable at the present level of precision. The framework is *fit-ready*: a single parameter can account for the leading correlated deviations from ΛCDM across more than ten orders of magnitude in scale.

Forecasts for upcoming experiments indicate that the next decade will decisively test the relaxation-driven, globally coherent structure of the Universe. Future CMB missions, large-scale structure surveys, and gravitational-wave observatories will measure α_{IR} with percent-level precision and probe the distinctive tensor-sector signatures predicted by RDCC. A consistent measurement of α_{IR} across all observational channels would provide strong evidence for the global CPT-symmetric architecture developed in Companions I–IV, while any significant mismatch would falsify the framework.

Companion V therefore completes the observational synthesis of RDCC and establishes the program as a unified, predictive, and empirically testable description of cosmic evolution. The relaxation-driven loss of global coherence, quantified by α_{IR}, emerges as the single parameter linking the earliest moments after the bounce to the largest observable scales of the Universe.

Appendix A — Minimal Python Implementation of the RDCC Tensor Spectrum

This appendix provides a minimal Python implementation of the RDCC tensor spectrum, including the characteristic log-periodic modulation arising from interference between the two CPT-related tensor sectors across the bounce. The code is intended as a lightweight reference implementation for reproducing the qualitative features of the RDCC stochastic gravitational-wave background (SGWB). A full numerical pipeline, including likelihood evaluation and parameter inference, is provided in the accompanying Companion Papers.

```
import numpy as np

# RDCC tensor spectrum with log-periodic modulation
def Omega_GW(k, alpha=0.017, eps=0.1, omega=3.2, kb=1e-3, phi=0.0):
    """
    Minimal RDCC tensor spectrum.
    Parameters:
        k     : comoving wavenumber
        alpha : relaxation parameter
        eps   : modulation amplitude
        omega : logarithmic frequency
        kb    : bounce scale
        phi   : phase shift
    """
    # Blue-tilted background spectrum
    Omega0 = k**(2*alpha)

    # Log-periodic modulation
    modulation = 1.0 + eps * np.cos(omega * np.log(k/kb) + phi)

    return Omega0 * modulation

# Example usage:
k = np.logspace(-4, 1, 500)
spectrum = Omega_GW(k)
```